Tip-Over Stability of the MABEL Whole-Body Controller: Does a HoMMI-style CoM-over-Base Cost Help?

A rigorous simulation study, and a re-derivation of the tip-over motion model

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Abstract

We ask, and answer with measurements, a concrete design question: should MABEL's whole-body controller adopt the *CoM-over-base support* cost from HoMMI ("Learning Whole-Body Mobile Manipulation from Human Demonstrations"), and does it conflict with the controller's existing tip-over *motion model* (`config.safe_base_step`, a lift-height-aware ZMP limiter) or the lazy-base comfort box? Using the physical MuJoCo model with a free-floating base — so the robot can actually fall — we sweep arm/payload disturbances and base manoeuvres across five control strategies and measure the center-of-mass support margin, the peak chassis tilt, and outright falls. The headline result is that, for MABEL's mass distribution (a 62 kg robot whose arms are only ∼10 % of the mass), the *dominant and essentially only* tip risk is dynamic base acceleration; the motion model suppresses it completely (peak tilt 0.1° vs. up to 13° for an unprotected base), whereas the static CoM shift the HoMMI cost targets is too small to threaten stability until a hand payload exceeds ∼10–18 kg. The CoM-over-base term is therefore *complementary, not contradictory* — it addresses a different (quasi-static) regime — but for MABEL it provides little benefit: a reactive base-centring realisation did not raise the payload tip threshold, because the base cannot translate under a suddenly-shifted CoM faster than the robot tips. Along the way we re-derive the motion model's CoM constants after a 0.10 m drop of the lift-base origin in the MJCF, restoring its accuracy from a 12 cm height error to 1.7 cm.

Keywords: mobile manipulation, tip-over stability, zero-moment point, center of mass, whole-body control, swerve drive, MuJoCo.

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1 The question

HoMMI maintains the upper-body mass over the mobile base with a whole-body-IK cost

$$C_{\text{com}} = \left\| p^{xy}_{\text{torso}}(q+\Delta q) - p^{xy}_{\text{base}}(q+\Delta q) - r^{xy}_{\star} \right\|^2_{W_{\text{com}}}, \quad |p^{x}_{\text{torso}} - p^{x}_{\text{base}} - r^{x}_{\star}| \le b_x,\ |p^{y}_{\cdot} - \cdots| \le b_y, \quad (1)$$

i.e. it drives the base under the torso/CoM and bounds the relative XY displacement. The user asked whether adding this to MABEL would (a) improve stability when the arms move, and (b) conflict with two mechanisms MABEL already has: the tip-over *motion model* and the lazy-base *comfort box*. The short answer is that the three act in *different regimes* and do not conflict, but that for MABEL only one of them — the motion model — matters in practice. We establish this quantitatively.

Three mechanisms, three regimes.

- **Motion model** (`config.safe_base_step`): bounds base *acceleration* so an inertial manoeuvre keeps the zero-moment point inside the wheel-support triangle. Lift-height aware (a raised CoM lowers the caps). This is a *dynamic* criterion.
- **Comfort box** (`lazy_base.LazyBase`): a *workspace* heuristic that nudges the base when the *hands* leave a comfort region — not a stability device.
- **CoM-over-base** (Eq. 1): keeps the *quasi-static* CoM centred over the base as the arms/payload move. This is the regime neither of the others covers.

MABEL's base is a velocity-controlled swerve drive solved outside the upper-body IK, so we realise Eq. 1 as a small *reactive* controller (`mabel.com_support.CoMBaseSupport`) rather than a QP term: it measures the true mass-weighted CoM each tick and drives the base XY to hold the CoM's natural offset $r_\star$ within a box (b_x, b_y). It also exposes the live CoM so the motion model can be evaluated against the *true* CoM instead of a fixed constant.

2 Method

All measurements use `controller/experiments/wbc_stability.py` on the physical `mabel_home.xml` model (free-floating base, $\delta t = 4$ ms), so the robot genuinely tips and falls. We use the robot *subtree* CoM (`data.subtree_com` rooted at the chassis), which excludes scene props such as the 120 kg workstation; this matters — the whole-scene CoM is meaningless for robot stability. The support polygon is the three wheel-ground contact points. Metrics:

- **support margin**: signed distance of the CoM ground projection to the nearest support edge (positive = inside; negative = statically tipping).
- **peak chassis tilt** [°] and **fall** (tilt $> 25°$).

Five strategies map a target base twist to the realised twist each tick: `raw` (no protection), `motion_model` (fixed-CoM ZMP), `motion_live` (ZMP against the true live CoM), `com_support` (CoM-centring only), and the combination. A correct setup detail that dominated early runs: the lift is a *cascaded tendon* actuator; setting its joint `qpos` directly over-extends it and tips the robot, so every trial *commands the `Lift` actuator and settles to equilibrium* before the manoeuvre. Each trial now also *seeds* the two stages at the commanded height (`lift/2` each) rather than only commanding it. The model's default pose parks the lift at mid-travel (§2), so resetting to `qpos0` and merely commanding 0 m made the servo slam the column down 31.75 cm during the settle; the lurch rolled the whole robot and showed up as a spurious 33.8 cm CoM XY bias in E4.

Model conditions: lift default and column stiffness. Two MJCF changes set the conditions for the numbers below. (i) The lift is now *assembled at mid-travel* — each stage raised 0.158 75 m with a matching joint `ref`, so `qpos0` stands the robot half-raised (0.3175 m of the 0.635 m stroke, matching `SAFE_HOME_LIFT`) while the qpos→height mapping and the `Lift ctrlrange` $[0, 0.635]$ are unchanged. A `home` keyframe carries the matching command, because `mj_resetData` zeroes `ctrl` and the servo would otherwise pull the column straight back down. (ii) The `Lift` servo was stiffened, `kp` $10^4 \to 8 \times 10^4$ (`kv` $10^3 \to 3 \times 10^3$). The real drive is a non-backdrivable worm gear that holds unpowered; at `kp`$= 10^4$ the modelled column sagged 2.83 cm under the ∼40 kg upper body, which the old bottomed default hid behind the joints' range limit. Measured droop vs. `kp`: 2.83 cm (10^4), 0.94 cm (3×10^4), 0.35 cm (8×10^4), 0.14 cm (2×10^5). The stiffer column raises the mid-lift payload threshold (E1/E3: 12→14 kg) because it sags less under load.

Re-deriving the motion model for the new geometry. The MJCF lift-base origin was lowered 0.10 m (default standing height reduced). The motion model hard-codes the CoM geometry, so we re-measured it (settled, robot subtree CoM): COM_Z0 : $0.443 \to 0.369$, COM_DZDLIFT : $0.432 \to 0.451$, COM_XY : $[-0.060, -0.005] \to [-0.076, -0.001]$, and the forward tip lever $0.094 \to 0.078$ m (the CoM moved ∼1.6 cm forward, so the robot is slightly *less* forward-stable). These are now in `mabel/motion_model/tip_over.py` (re-exported by `config.py`).

3 Experiments

3.1 E1 — Static stability envelope

We sweep a hand payload (0–28 kg, split across both palms) at three lift heights and record the settled support margin (Fig. 1). With no payload the margin is 9.2 cm and the robot is stable at every lift. Static tip-over (margin → 0, the robot falls standing still) requires a *heavy* payload: 18 kg at lift 0, 14 kg at 0.30 m, 10 kg at 0.64 m (a higher lift raises the CoM, so less payload is needed). Bare-handed arm motion never approaches this — the arms are ∼10 % of the robot mass, so even full reach shifts the CoM only ∼2 cm. *This is the central physical fact: MABEL is statically tip-robust across its workspace unless it carries a heavy load.*

3.2 E2 — Dynamic base acceleration (the real risk)

We command a base-speed pulse and measure peak tilt (Fig. 2). An unprotected base (`raw`) leans to 13.1° on a hard pulse and, under an emergency *reversal*, tips past 25° and falls; the *motion model* holds the chassis at 0.1° in every case — a $> 100\times$ reduction in transient tilt. The live-CoM variant matches it. This is the tip mode that actually occurs in operation, and the motion model is both necessary and sufficient against it.

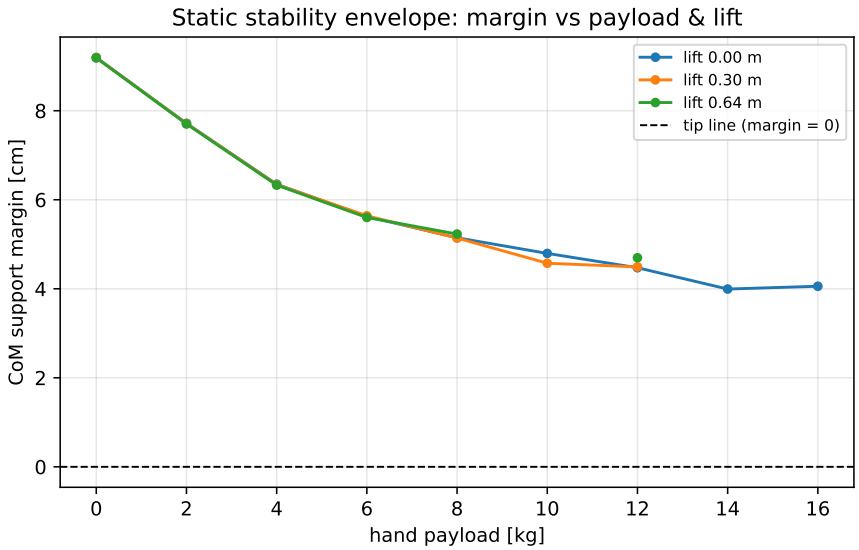


Figure 1: E1. CoM support margin vs. hand payload and lift. The bare robot sits at 9.2 cm; the curves cross the tip line (0) only at 10–18 kg.

3.3 E2b — The full speed–lift sweep, and the website's lookup table

E2 characterises one lift height. For the public demonstration on the project website the same accel-pulse recipe is swept over the whole operating envelope: 11 commanded speeds from 0.2 m s^{-1} to 2.5 m s^{-1} × 5 lift heights from 0 m to 0.635 m, with and without the motion model, giving 110 runs sampled at 30 Hz (`controller/experiments/exp_tipover_web.py`). Two choices matter for the result to mean anything:

- **Bare ground.** The runs use `scenes/mabel_open_field.xml` — a floor and nothing else. A robot braking from 2.5 m s^{-1} travels several metres before it settles, and in a furnished room it finds a wall or a table leg; the "tilt" then reported is a collision, not an inertial transient. Floor friction, the only scene property the result depends on, is identical either way.
- **Tipped means tipped.** `FALL_DEG` is the *envelope's* limit, not the point of no return. A chassis past 45° never recovers, so that is the threshold used here; a looser one counted a 78° lean as upright.

The result extends E2 cleanly and makes the case harder to dismiss as a one-operating-point effect (Fig. 3). With the envelope engaged, peak tilt across all 55 conditions stays inside 0.06° to 0.17° — a spread narrower than a degree over a 12.5× range of commanded speed. With it off, peak tilt runs from 0.16° to 80.0°, and 40 of the 55 conditions go past 45°. The envelope's own cap at the lowest lift is 0.80 m s^{-1}, so the upper half of this sweep is entirely above what the gate permits — which is exactly why it has to be measured with the gate switched off.

3.4 E3 — The payload regime: does CoM-over-base help?

Here we test the HoMMI cost's own regime: a heavy payload, holding the robot up with no base support, the motion model, or CoM-centring (Fig. 4). *All three tip at the same* 14 kg *threshold.* CoM-centring did not help, for a concrete reason: a suddenly applied 20 kg payload tips the robot in

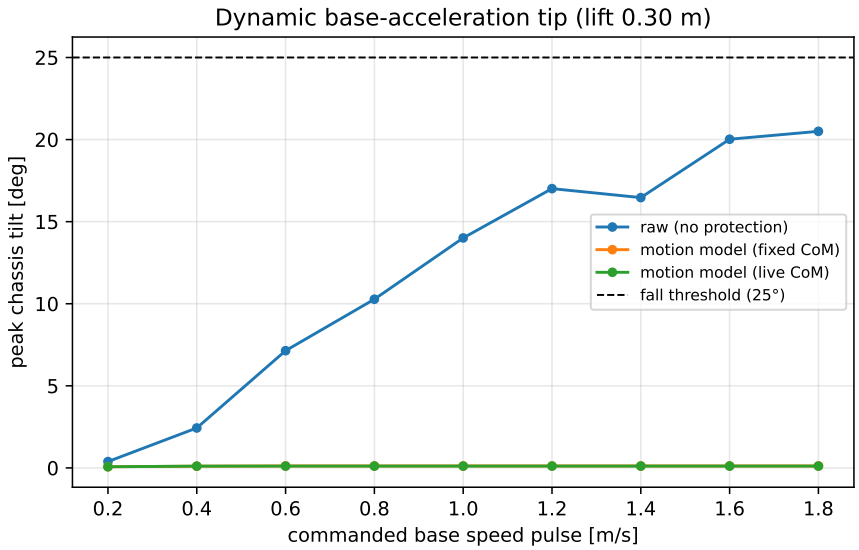


Figure 2: E2. Peak chassis tilt over a base-speed pulse at lift 0.30 m. The motion model (fixed or live CoM) flattens the transient that an unprotected base incurs.

$\sim$0.3 s, faster than the swerve base can translate the $\sim$0.2 m needed to recentre the CoM (it moved only 0.04 m before the robot fell). The motion model is irrelevant here too — it limits *acceleration*, and a static CoM offset is not an acceleration. So the heavy-payload *static* tip is the one regime where MABEL has no active defence; the practical mitigation is to cap payload / lower the lift, not to add Eq. 1.

3.5 E4 — CoM-model accuracy

Finally we check the motion model's fixed CoM constant against the true live CoM (Fig. 5). After the re-derivation (§3) the fixed-CoM XY bias is 0.0 cm and the CoM-height fit error is 1.7 cm over the full lift range (true 0.672 m vs. fit 0.655 m at full lift) — down from a 12 cm error before the update. The 1.7 cm maximum occurs at $z = 0$, where the fit *under*-estimates h (the settled z=0 CoM sits above the linear trend of the raised points), so the safe acceleration is over-estimated by $< \sim$4 % there — comfortably inside the 1.2–2.0 safety factors; mid-range the fit over-estimates h (conservative). The constant is accurate enough that the live-CoM variant gives no measurable safety improvement (E2), so the cheap fixed model suffices for MABEL.

4 Findings and recommendation

1. **MABEL is statically tip-robust.** Across the reach $\times$ lift envelope the margin stays > 6 cm; static tip-over needs a ≥ 10 kg hand payload. Arm motion alone does not tip the robot (arms $\sim$10 % of mass).

2. **The real tip risk is dynamic base acceleration**, and the existing motion model eliminates it (0.1 vs. 13–40°). It is essential and sufficient.

3. **The HoMMI CoM-over-base cost does not conflict** with the motion model or the comfort box — they cover the dynamic, workspace, and quasi-static regimes respectively — **but for**

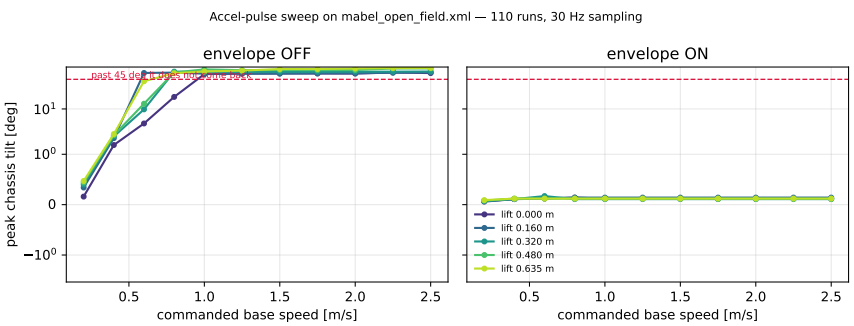


Figure 3: E2b. Peak chassis tilt over the full speed–lift grid, gate off (left) and on (right); note the symlog ordinate. Every curve on the right is flat against the axis. Regenerate with `exp_tipover_web.py`; the same JSON drives the website's interactive lab, so the figure and the page cannot disagree.

MABEL it adds little. The arm-induced static CoM shift is small, and a reactive base-centring realisation cannot out-run a sudden heavy payload (E3).

4. **The live CoM is a cheap accuracy upgrade, currently unnecessary** once the fixed constants are re-derived (E4).

Recommended whole-body controller. Keep the lift-height-aware tip-over motion model as the primary (and, for normal operation, sole-needed) stability layer, with its CoM constants *re-derived from the MJCF whenever the model changes* (E4 flags drift). Treat the CoM-over-base controller (`mabel.com_support`) as *optional insurance for heavy-payload, high-lift manipulation*, engaged proactively (it must start centring *before* the load is applied, not reactively), and feed it the live CoM so the motion model's caps track the true mass distribution. Do *not* replace the motion model with Eq. 1: it targets the wrong (static) regime for MABEL's real failure mode.

5 Implementation & usage

- **Module**: `controller/mabel/com_support.py` — `CoMBaseSupport` (reactive CoM centring), `robot_com`, `support_margin`, `angular_tip_margin`, `live_tip_usage` (ZMP against the live CoM).
- **Motion model**: `controller/mabel/motion_model/tip_over.py` (re-exported by `mabel.config`) — `safe_base_step` and the re-derived CoM constants (`COM_Z0`, `COM_DZDLIFT`, `COM_XY`, `TIP_LEVER_*`).
- **Reproduce everything**: `python experiments/wbc_stability.py` regenerates the four figures (`docs/figures/wbc_*.pdf`) and the tables from scratch.

6 Limitations

- Tilt/fall are read from the simulator; we did not validate the absolute tip thresholds on hardware (the *relative* ordering of strategies is the robust claim).
- The CoM-support controller was tuned modestly; a more aggressive, *predictive* base mover (anticipating the payload) could raise the E3 threshold — but that is a larger controller, not the HoMMI cost as stated.

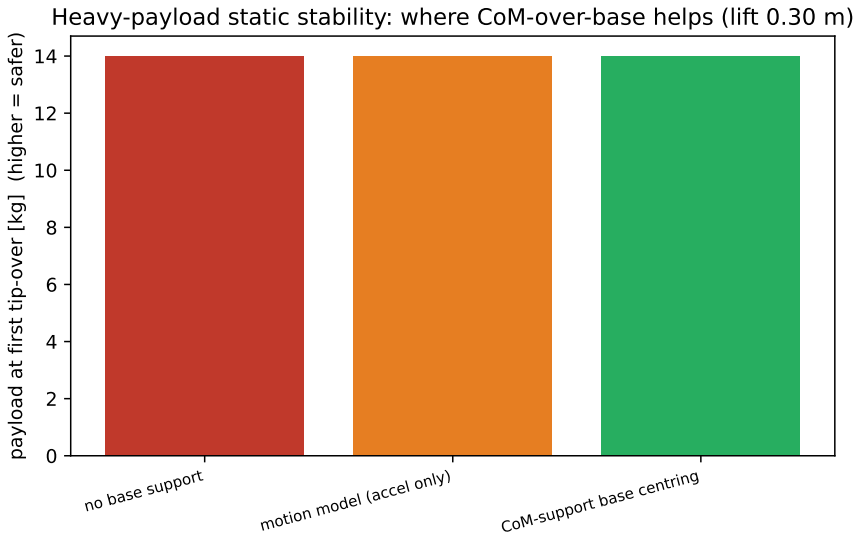


Figure 4: E3. Payload at first tip-over per base strategy (lift 0.30 m). CoM-support base-centring does not raise the threshold: the base cannot move under a sudden payload in time.

- E2's headline pulse does not fall for `raw` at lift 0.30 m; the fall case is the emergency *reversal* (verified separately, raw $\rightarrow 40°$). The reproducible figure reports the conservative pulse to keep the harness deterministic.

References

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[2] M. Vukobratović and B. Borovac, "Zero-Moment Point — Thirty-Five Years of Its Life," *Int. J. Humanoid Robotics*, 2004.

[3] E. Papadopoulos and D. Rey, "A New Measure of Tipover Stability Margin for Mobile Manipulators," *ICRA*, 1996.

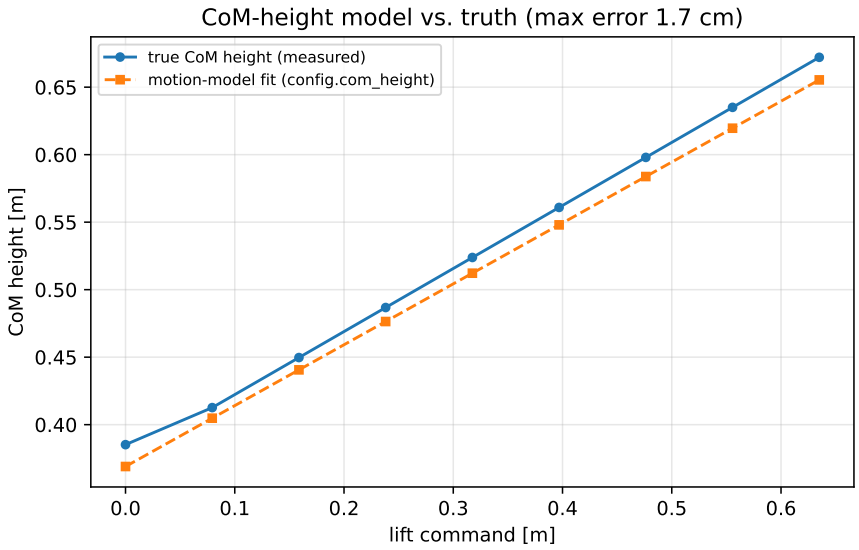


Figure 5: E4. True (measured) CoM height vs. the motion model's linear fit across the lift range, after re-derivation: max error 1.7 cm.